AI – TOOLS
&
EFFECTIVE PROMPT PRACTICES

Prompt comparision GPT/Gemini/Claude

- **Explain climate change**

- **Analyze**
- Which answer is easiest to understand?
- Which gives the best structure?
- Which uses examples?
- Which answer is too long?

gpt

Explain climate change.

Climate Change

Climate change refers to long-term changes in Earth's temperature, rainfall, wind patterns, and other climate conditions. While Earth's climate has naturally changed throughout history, the rapid warming observed in recent decades is primarily caused by **human activities**.

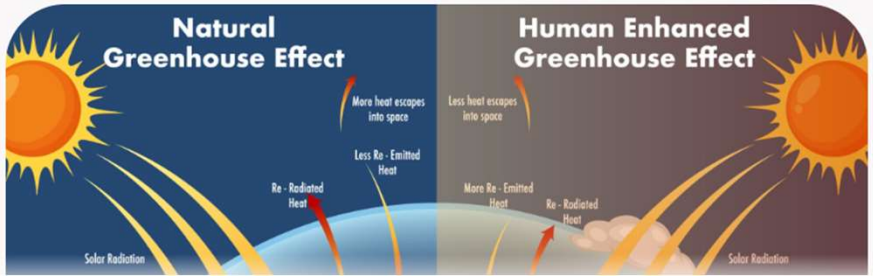
The main cause is the increasing concentration of **greenhouse gases** such as carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O). Burning coal, oil, and natural gas for electricity, transportation, and industry releases large amounts of CO_2. Deforestation also contributes because forests absorb CO_2 from the atmosphere. These gases trap heat, strengthening the **greenhouse effect** and causing global temperatures to rise.

Climate change can lead to **more frequent and intense heatwaves, changing rainfall patterns, droughts, floods, melting glaciers and ice sheets, rising sea levels, and disruptions to ecosystems and agriculture**. Its effects vary by region, and some communities are more vulnerable than others.

gemini

Explain climate change.

Climate change refers to long-term shifts in global temperatures and weather patterns. While natural factors like volcanic eruptions or solar cycles historically shifted Earth's climate, human activities since the 1800s have become the primary driver.



claude

Explain climate change.

Climate change refers to long-term shifts in temperatures and weather patterns, primarily driven by human activities since the mid-1800s, mainly the burning of fossil fuels (coal, oil, and gas).

The basic mechanism

Earth's atmosphere naturally contains gases—carbon dioxide, methane, water vapor, and others—that trap heat from the sun. This "greenhouse effect" is essential for life; without it, Earth would be far too cold. The problem arises when human activities add extra greenhouse gases to the atmosphere, intensifying this effect and trapping more heat than the planet naturally would.

Main sources of emissions

- Burning fossil fuels for electricity, heat, and transportation (the largest source)
- Deforestation, which reduces the planet's ability to absorb CO2
- Agriculture, particularly livestock farming (methane) and fertilizer use (nitrous oxide)
- Industrial processes like cement and steel production

Prompt Anatomy

- Role + Task + Context + Constraints + Output Format + Examples

```
You are an experienced engineering faculty member.

Task:
Explain machine learning to first-year engineering students.

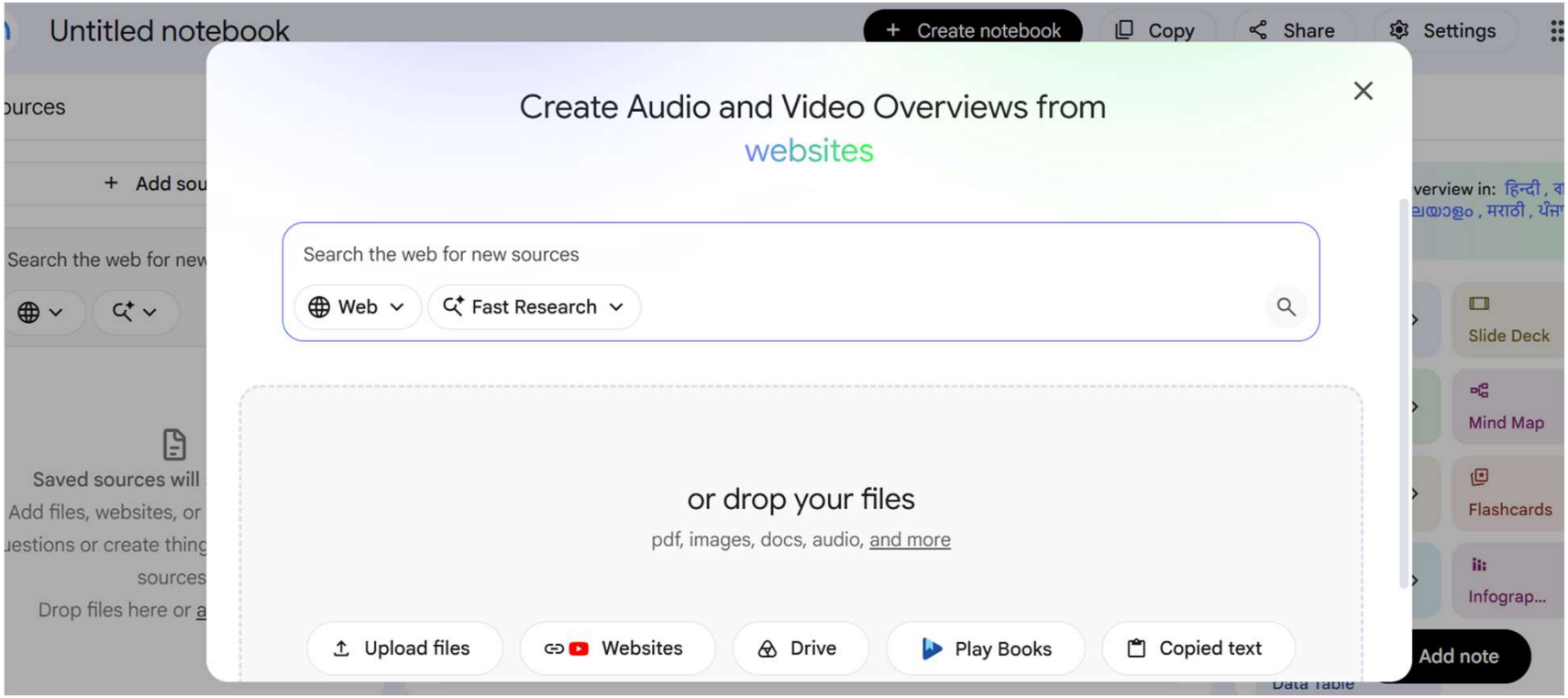
Context:
The students have basic Python knowledge but no ML background.

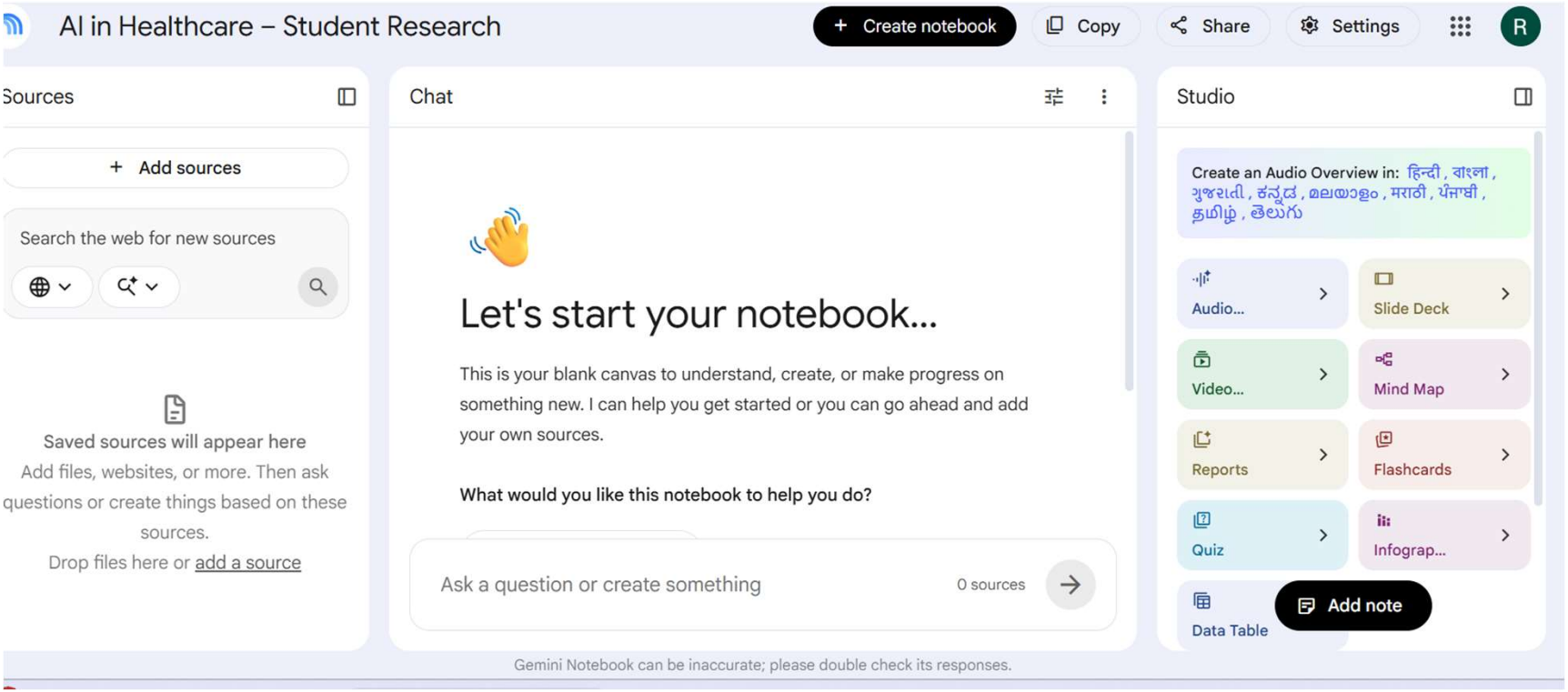
Requirements:
- Use simple language
- Explain with a real-world example
- Explain supervised and unsupervised learning
- Give 3 applications

Output:
Use headings and bullet points.
Keep the answer below 500 words.
```

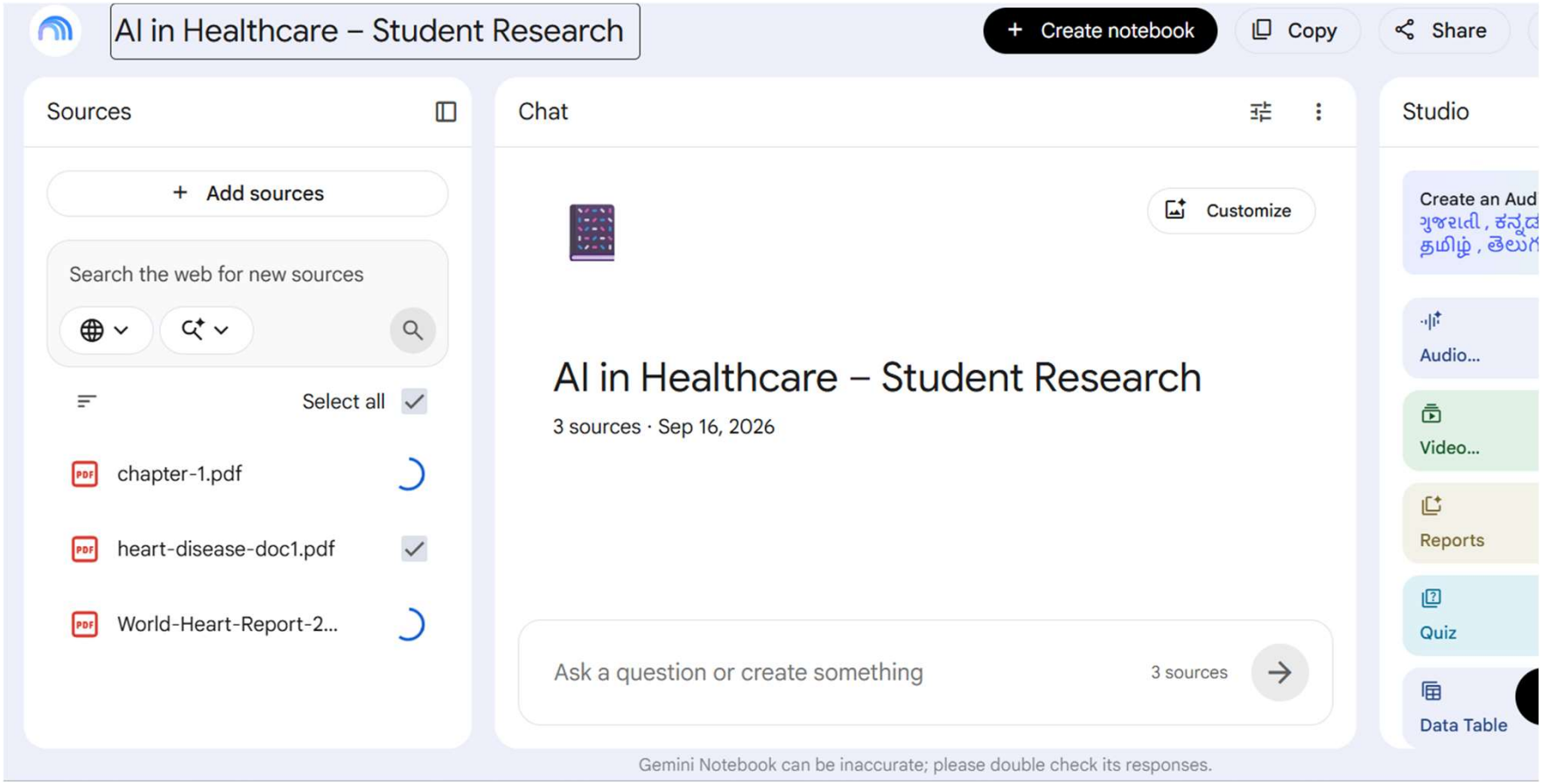
- **what is NotebookLM?**
- **NotebookLM** is Google's AI research and learning assistant.
- Instead of asking an AI to answer only from its general knowledge, you can provide your own sources—such as PDFs, documents, websites, slides, or other supported materials—and ask questions about those sources.

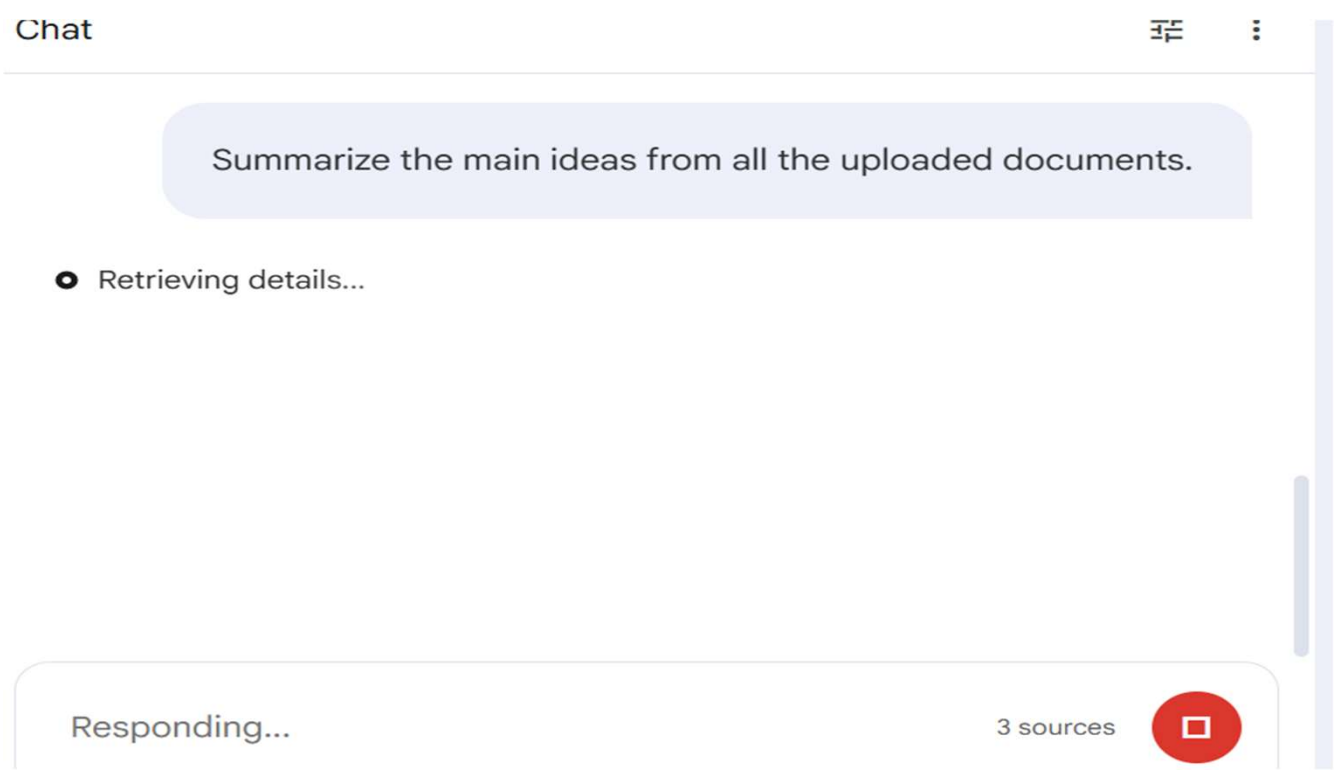
Note Book LM - google




AI in Healthcare – Student Research
+ Create notebook
Copy
Share
Settings
R
Sources
+ Add sources
Search the web for new sources
Saved sources will appear here
Add files, websites, or more. Then ask questions or create things based on these sources.
Drop files here or add a source
Chat
Let's start your notebook...
This is your blank canvas to understand, create, or make progress on something new. I can help you get started or you can go ahead and add your own sources.
What would you like this notebook to help you do?
Ask a question or create something
0 sources
Studio
Create an Audio Overview in: हिन्दी , বাংলা , ગુજરાતી , ಕನ್ನಡ , മലയാളം , मराठी , ਪੰਜਾਬੀ , தமிழ் , తెలుగు
Audio...
Slide Deck
Video...
Mind Map
Reports
Flashcards
Quiz
Infograp...
Data Table
Add note
Gemini Notebook can be inaccurate; please double check its responses.

- Add your sources.
- For example:
- Health care documents


AI in Healthcare – Student Research
+ Create notebook
Copy
Share
Sources
+ Add sources
Search the web for new sources
Select all
chapter-1.pdf
heart-disease-doc1.pdf
World-Heart-Report-2...
Chat
Customize
AI in Healthcare – Student Research
3 sources · Sep 16, 2026
Ask a question or create something
3 sources
Studio
Create an Aud
Audio...
Video...
Reports
Quiz
Data Table
Gemini Notebook can be inaccurate; please double check its responses.


Chat
Summarize the main ideas from all the uploaded documents.
Retrieving details...
Responding...
3 sources

Explain the difference between angina and a heart attack

- Looking at sources...

Responding... 3 sources

Ask different questions

- Teach me this document as if I am a beginner.
- Explain each major concept using a simple real-world example.
- Create 20 multiple-choice questions from this document.
- Include four options and the correct answer.

- Create a study guide from the uploaded material.

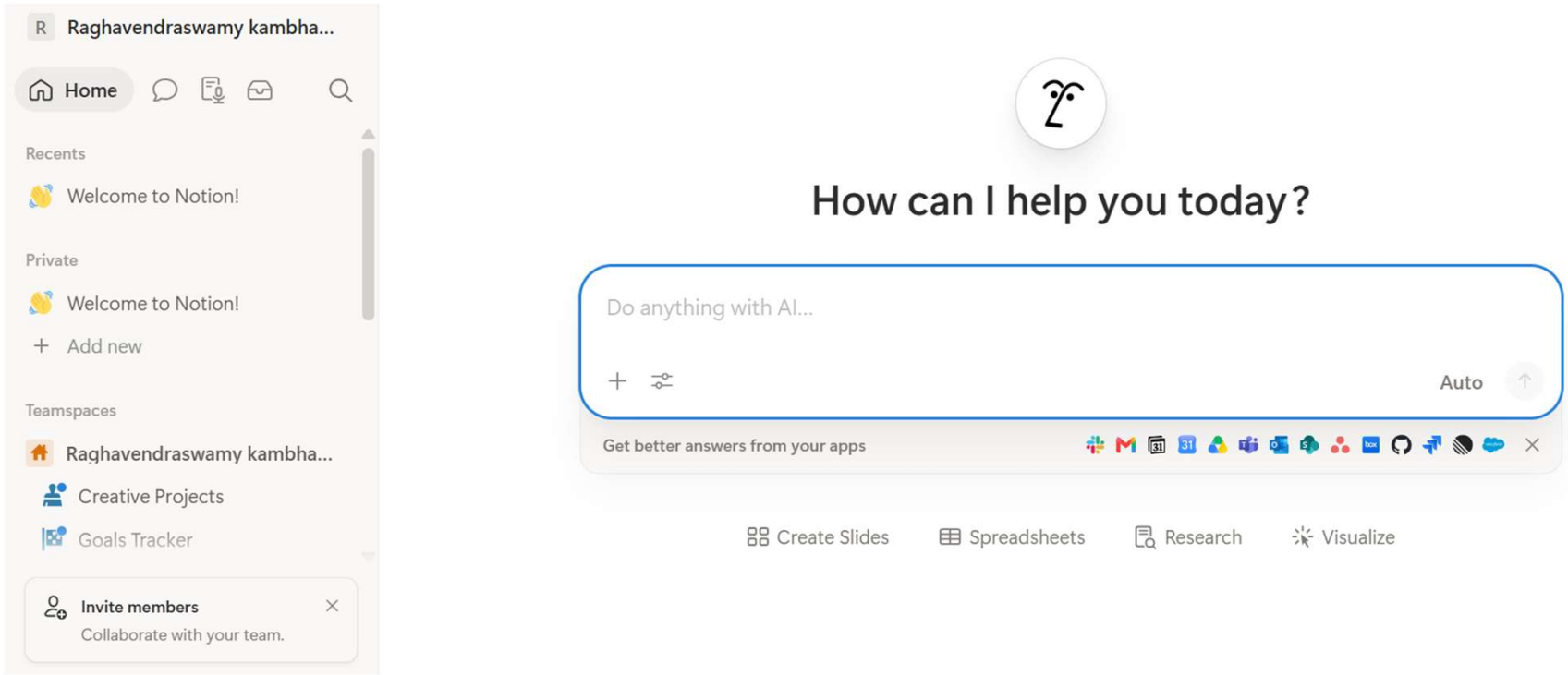
- Include:

- 1. Important definitions
- 2. Key concepts
- 3. Important formulas
- 4. Real-world examples
- 5. Common mistakes
- 6. Frequently asked questions
- 7. Examination questions
- 8. Quick revision points

Feature	What it does	Best use
Slide Deck	Converts your source material into a presentation	Teaching / seminars
Mind Map	Visually organizes concepts and relationships	Understanding a topic
Video Overview	Creates a video-style explanation from your sources	Visual learning
Quiz	Generates questions to test understanding	Practice / assessment
Infographic	Converts information into a visual summary	Posters / quick learning
Flashcards	Creates question–answer cards for revision	Exam preparation

https://app.notion.com/ai

- https://app.notion.com/ai


Raghavendraswamy kambha...
Home
Recents
Welcome to Notion!
Private
Welcome to Notion!
Add new
Teamspaces
Raghavendraswamy kambha...
Creative Projects
Goals Tracker
Invite members
Collaborate with your team.
How can I help you today?
Do anything with AI...
Auto
Get better answers from your apps
Create Slides
Spreadsheets
Research
Visualize

- Start a draft:
- Create a AI Learning Hub for beginners.
- Include the following sections:
- 1. What is Artificial Intelligence?
- 2. What is Generative AI?
- 3. What are Large Language Models?
- 4. ChatGPT 5. Claude
- 6. Gemini
- 7. Prompt Engineering
- For each topic provide: -
- Simple explanation –
- Real-world example
- Make the content suitable for first-year engineering students.

- Meeting Notes creation using AI command
- Prepare content calendar
- Explore notion ai different fetures

https://gamma.app/

- https://gamma.app/

Back

Create from template

Your prompt

Import content

Prompting tips

Develop a training session for new hires on effective meeting facilitation.

Focus on common pitfalls, core facilitation functions, and practical tips as outlined in this document. Tailor the content to be relevant for new employees, emphasizing skill development and practical application.

Markdown

5118/50000

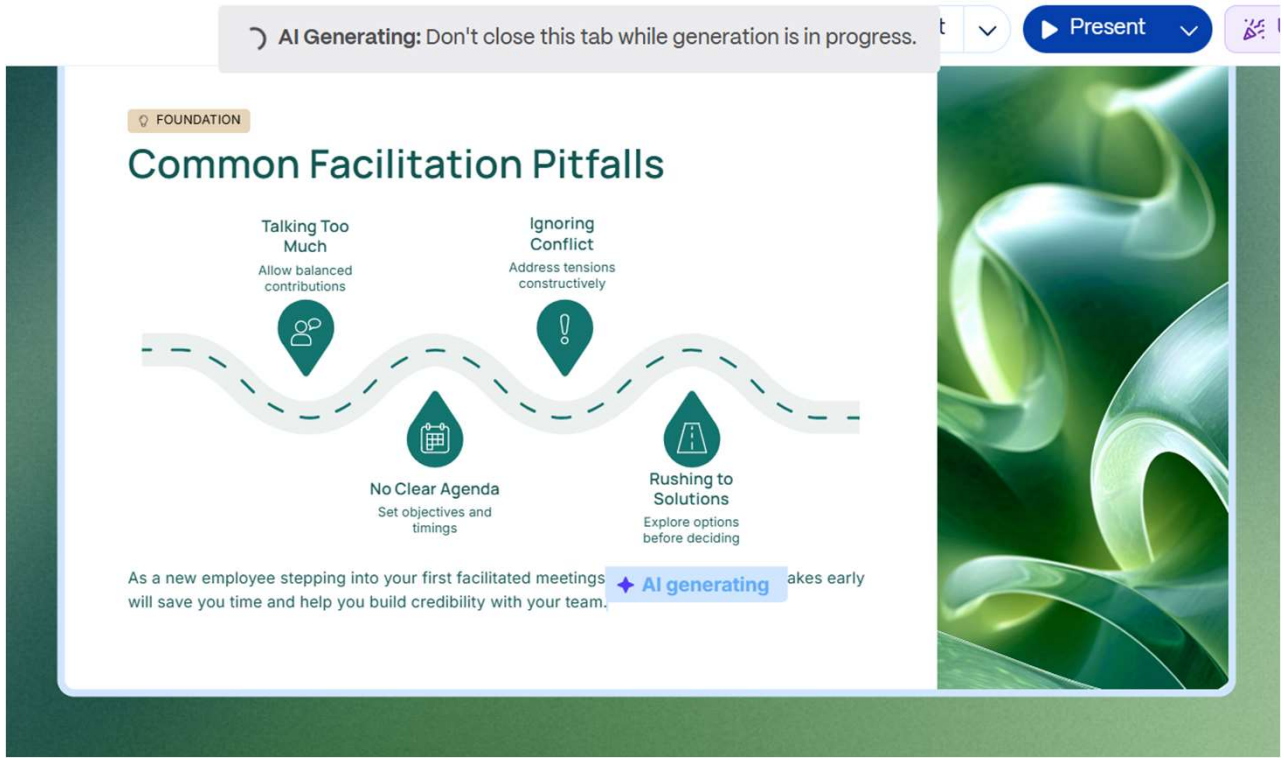
Visuals

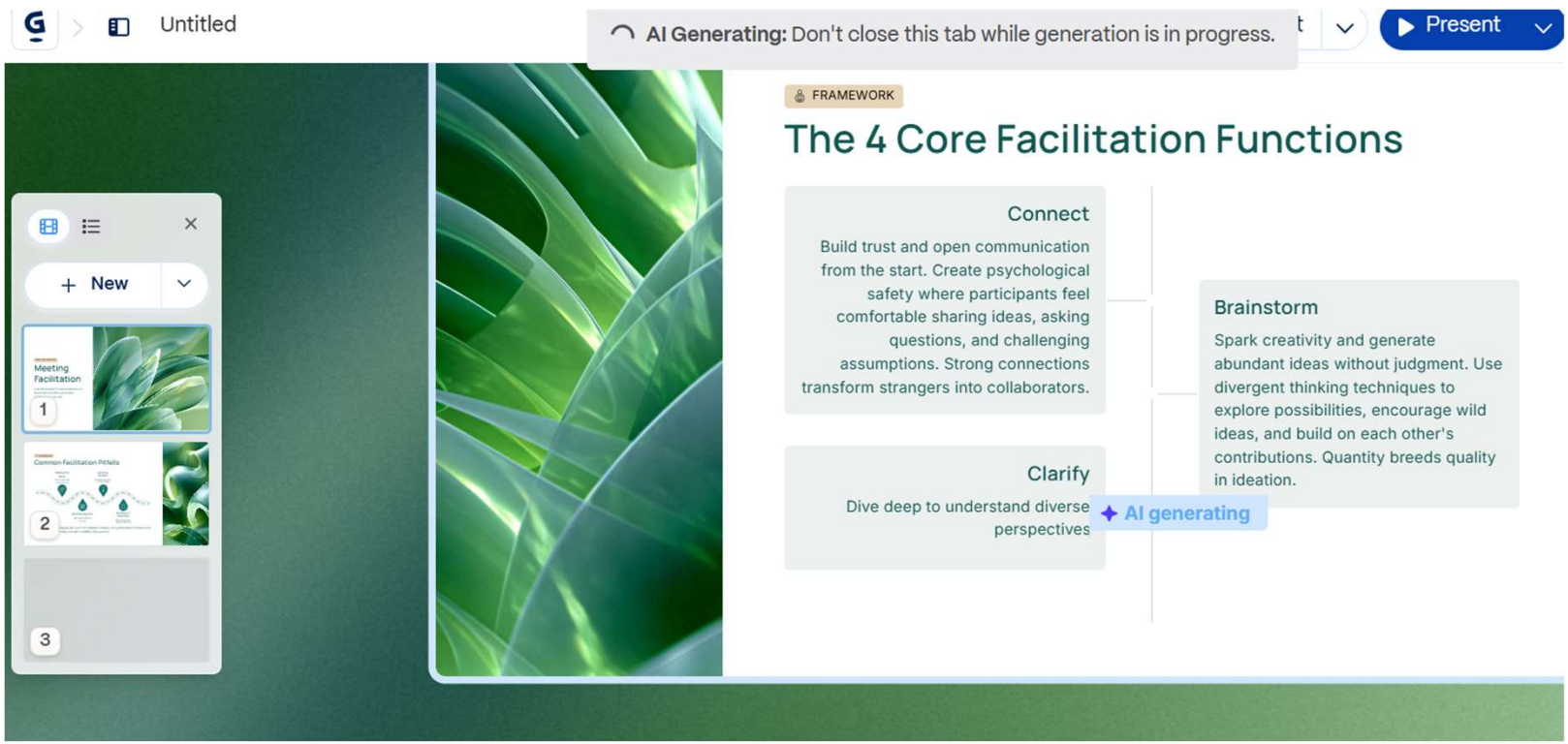
Accent images (selected theme has no accent images)

Tip: you can drag images you want to use directly into the prompt editor

Workshop Facilitation




AI Generating: Don't close this tab while generation is in progress.
Present
FOUNDATION
Common Facilitation Pitfalls
Talking Too Much
Allow balanced contributions
Ignoring Conflict
Address tensions constructively
No Clear Agenda
Set objectives and timings
Rushing to Solutions
Explore options before deciding
As a new employee stepping into your first facilitated meetings
AI generating
akes early
will save you time and help you build credibility with your team.


Untitled
AI Generating: Don't close this tab while generation is in progress.
Present
New
Meeting Facilitation
1
2
3
FRAMEWORK
The 4 Core Facilitation Functions
Connect
Build trust and open communication from the start. Create psychological safety where participants feel comfortable sharing ideas, asking questions, and challenging assumptions. Strong connections transform strangers into collaborators.
Brainstorm
Spark creativity and generate abundant ideas without judgment. Use divergent thinking techniques to explore possibilities, encourage wild ideas, and build on each other's contributions. Quantity breeds quality in ideation.
Clarify
Dive deep to understand diverse perspectives
AI generating

https://www.napkin.ai/

- https://www.napkin.ai/

Close ×

What do you want to create today?



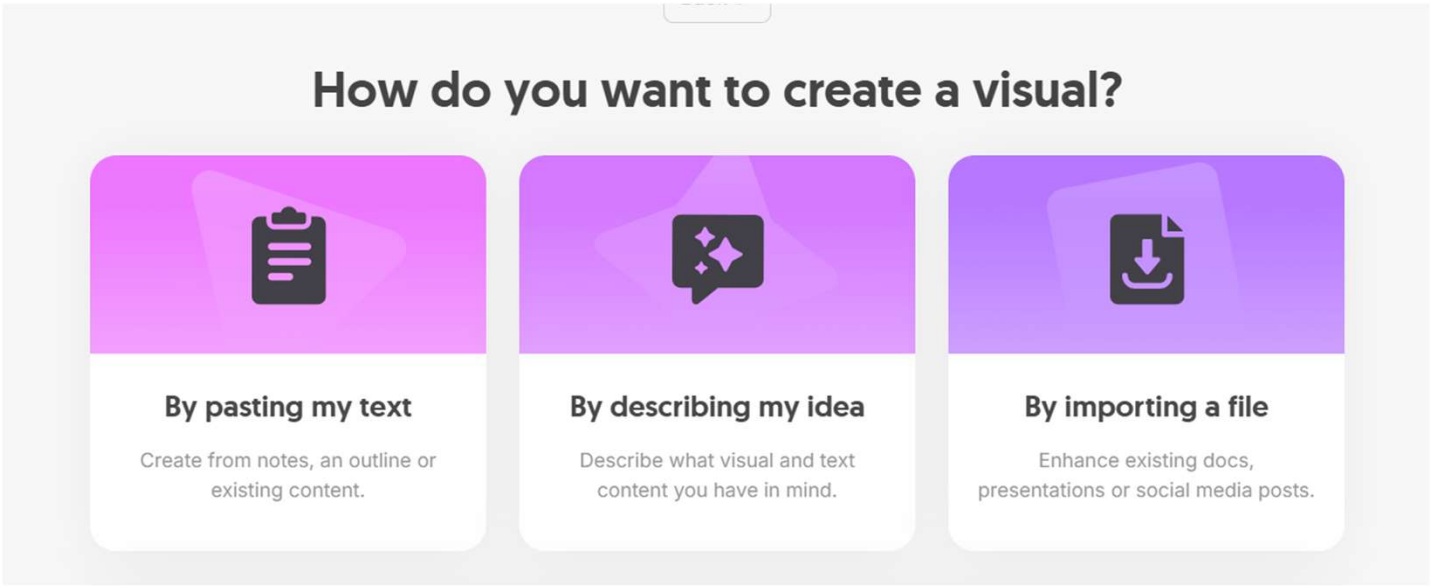
Visuals

Turn any text into a visual.



Slides BETA

Build a full presentation.


How do you want to create a visual?
By pasting my text
Create from notes, an outline or existing content.
By describing my idea
Describe what visual and text content you have in mind.
By importing a file
Enhance existing docs, presentations or social media posts.

Describing idea

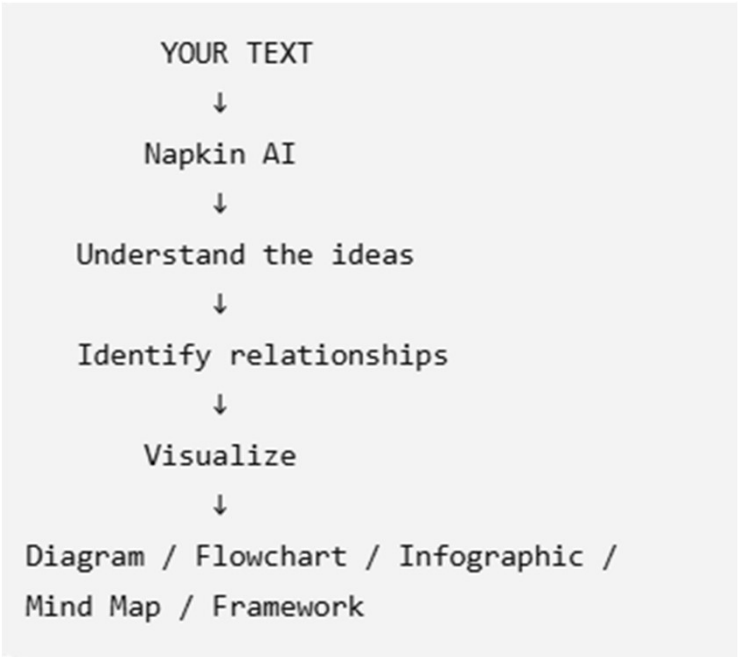
The Joyful Dance of the Peacock in the Rainy Forest

In the heart of a deep forest, where the lush greenery thrives and nature sings its melod tunes, a magical scene unfolds as rain begins to fall. This document explores the enchar sight of a peacock dancing joyfully amidst the raindrops, celebrating the beauty of natu and the vibrant life within the forest.

As the first drops of rain touch the earth, the forest awakens. The leaves glisten with moisture, and the air fills with the earthy scent of wet soil. The gentle patter of rain creat soothing symphony, harmonizing with the rustling leaves and distant bird calls. In this se environment, the peacock emerges, a symbol of beauty and grace.

The Peacock's Splendor

The peacock, with its iridescent feathers, is a sight to behold. Its plumage displays a my


YOUR TEXT
↓
Napkin AI
↓
Understand the ideas
↓
Identify relationships
↓
Visualize
↓
Diagram / Flowchart / Infographic /
Mind Map / Framework

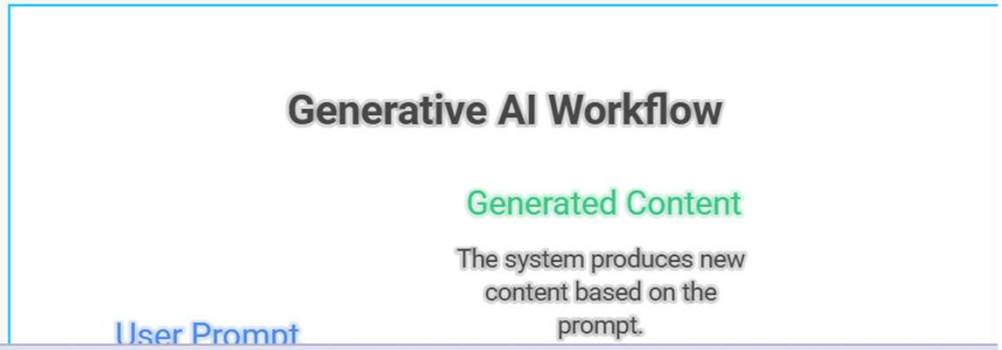
Generative AI is a branch of artificial intelligence that can create new content such as text, images, audio, video and code.

Large Language Models are AI models trained on large amounts of text. They can understand prompts and generate responses.

Generative AI applications include education, healthcare, finance, software development and content creation.

A typical Generative AI workflow is:

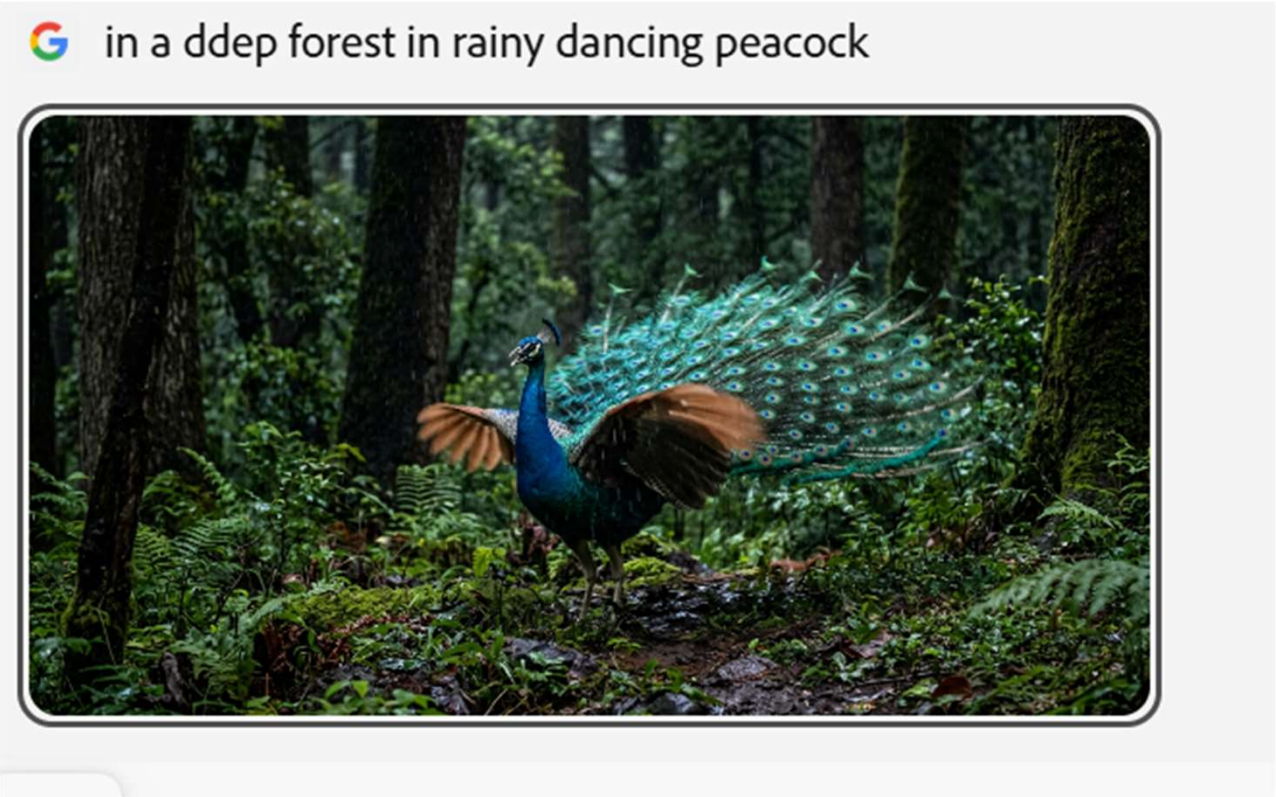
User Prompt → AI Model → Generated Content → Human Review.



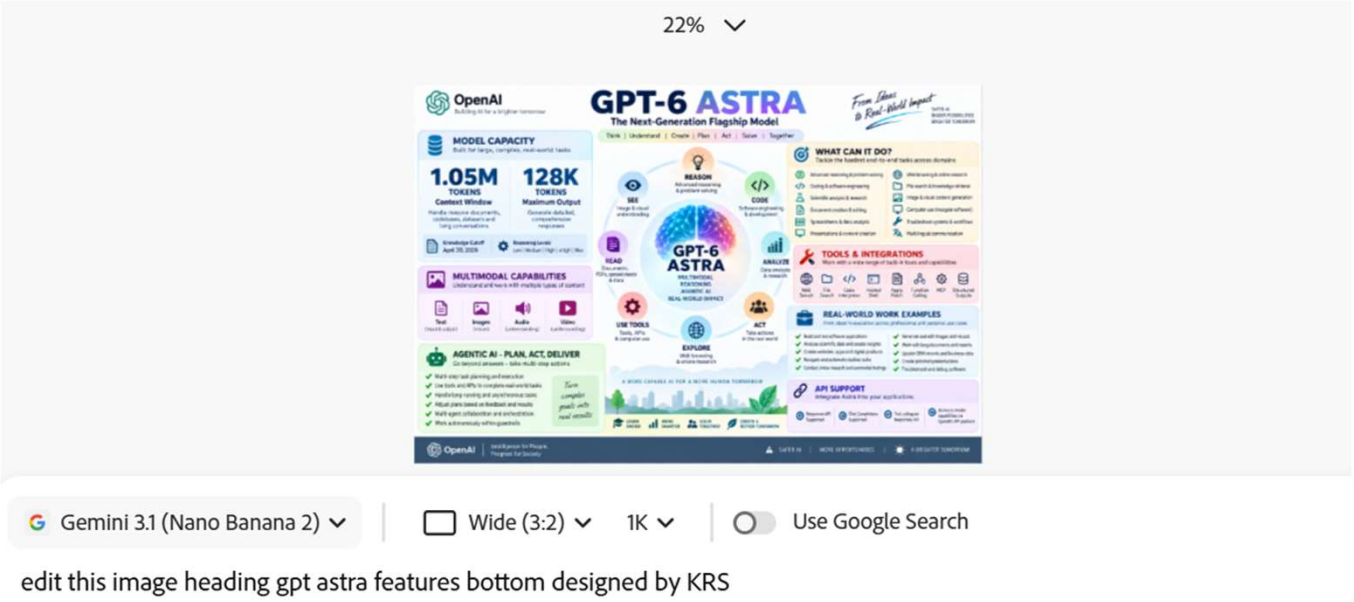
Give to your napkin create images

- Modern AI learning requires more than prompt engineering. Students should develop AI fundamentals, prompt engineering, critical thinking, data literacy, Python programming, API knowledge, RAG, AI agents, tool usage and responsible AI skills. They should also learn how to verify AI outputs and build real-world applications.

Adobe firefly create images

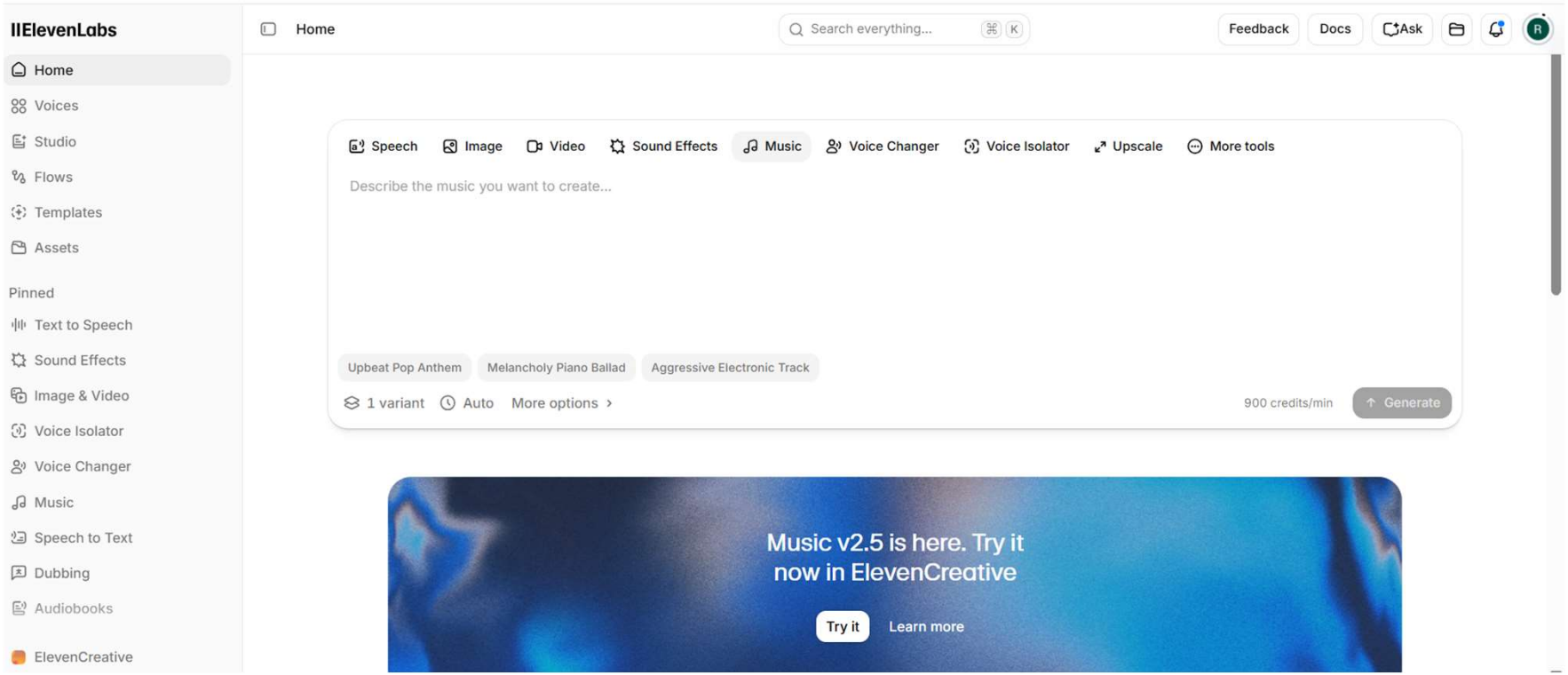


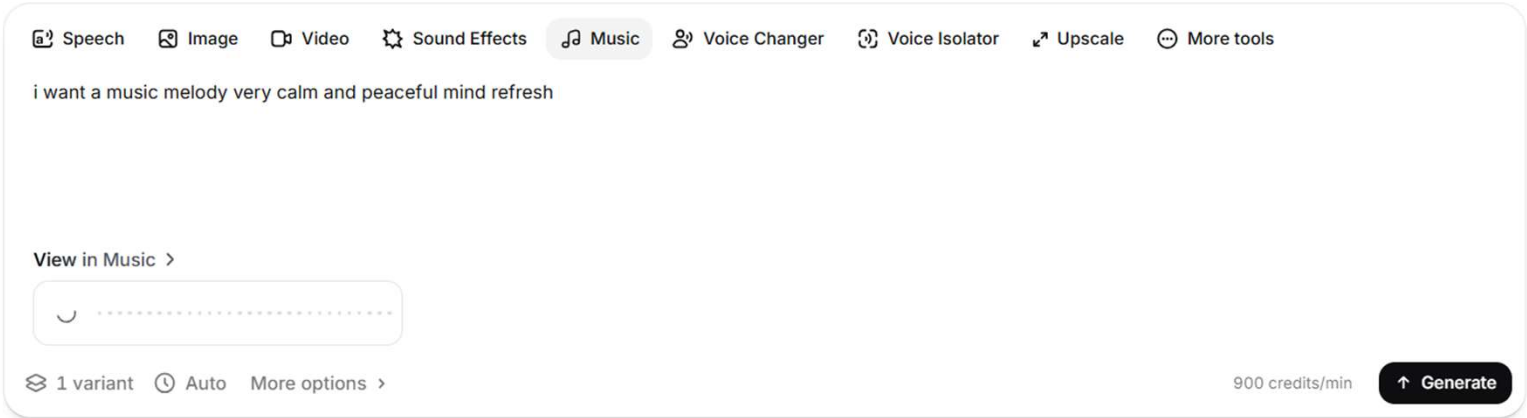
Edit images-adobe firefly

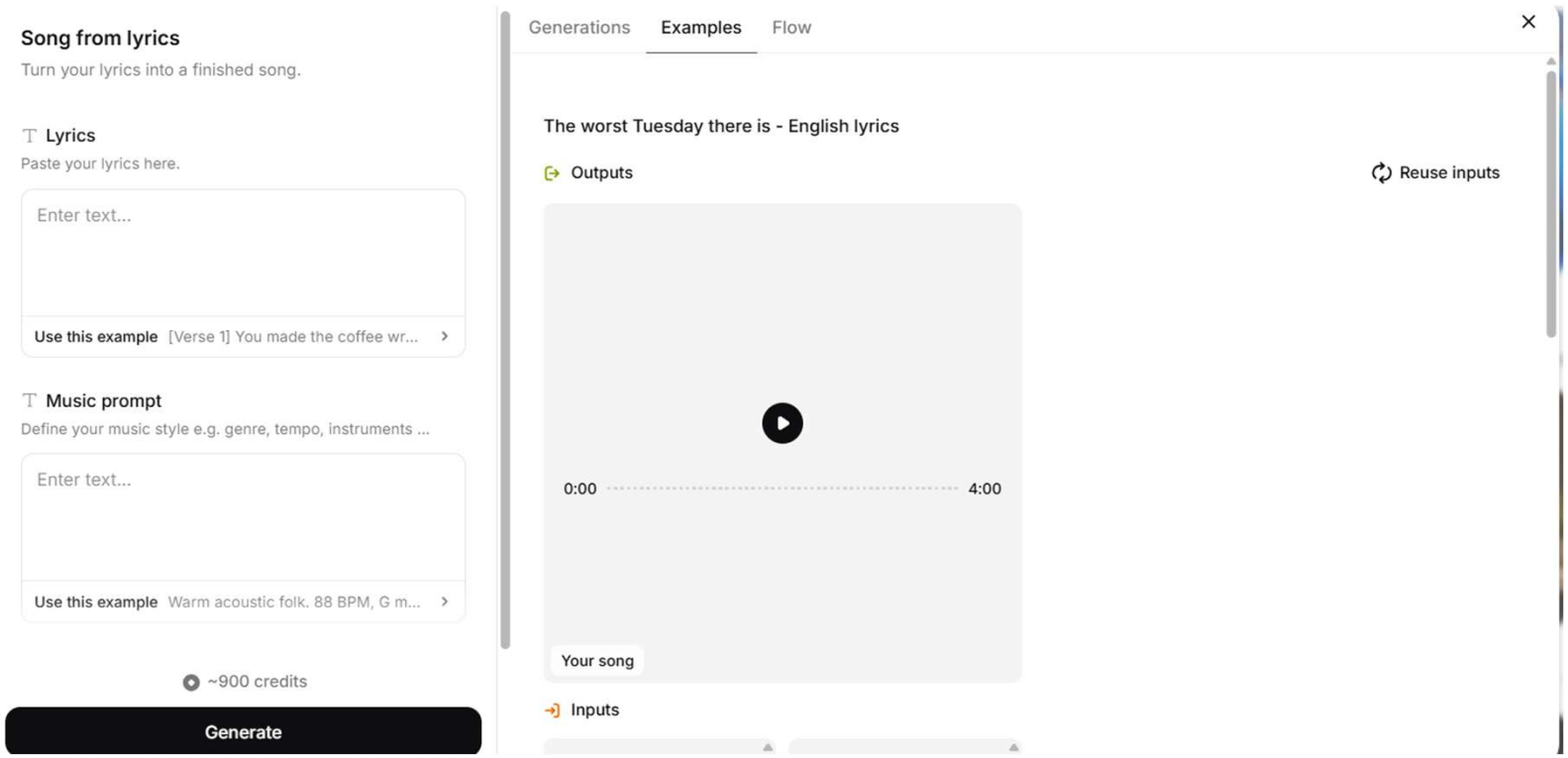


https://elevenlabs.io/app/home

- https://elevenlabs.io/app/home


IIElevenLabs
Home
Voices
Studio
Flows
Templates
Assets
Pinned
Text to Speech
Sound Effects
Image & Video
Voice Isolator
Voice Changer
Music
Speech to Text
Dubbing
Audiobooks
ElevenCreative
Home
Search everything...
Feedback
Docs
Ask
Speech
Image
Video
Sound Effects
Music
Voice Changer
Voice Isolator
Upscale
More tools
Describe the music you want to create...
Upbeat Pop Anthem
Melancholy Piano Ballad
Aggressive Electronic Track
1 variant
Auto
More options
900 credits/min
Generate
Music v2.5 is here. Try it now in ElevenCreative
Try it
Learn more


Speech
Image
Video
Sound Effects
Music
Voice Changer
Voice Isolator
Upscale
More tools
i want a music melody very calm and peaceful mind refresh
View in Music
1 variant
Auto
More options
900 credits/min
Generate


Song from lyrics
Turn your lyrics into a finished song.
Lyrics
Paste your lyrics here.
Enter text...
Use this example [Verse 1] You made the coffee wr...
Music prompt
Define your music style e.g. genre, tempo, instruments ...
Enter text...
Use this example Warm acoustic folk. 88 BPM, G m...
~900 credits
Generate
Generations
Examples
Flow
The worst Tuesday there is - English lyrics
Outputs
Reuse inputs
0:00
4:00
Your song
Inputs

- Rai rai raa raa
- Rai rai raa raa
- Adde leni daanai raa raa

- Rai rai raa raa
- Rai rai raa raa
- Adde leni dheenay raa raa

- High-energy Telugu folk-cinematic song with a powerful male lead
- vocal and energetic group chants. Fast tempo around 125–135 BPM.
- Prominent dappu, thappeta, hand percussion and deep bass drums,
- with nadaswaram phrases between vocal sections. Start with a
- short percussion intro, build into a catchy chorus, add call-and-
- response vocals, then progressively increase the percussion and
- energy toward a dramatic climax. Raw village-folk character
- combined with modern cinematic production. Festive, energetic,
- mass-celebration mood.

Song from lyrics

Turn your lyrics into a finished song.

Lyrics

Paste your lyrics here.

Adde leni daanai raa raa

Rai rai raa raa
Rai rai raa raa
Adde leni dheenay raa raa

Use this example [Verse 1] You made the coffee wr...

Music prompt

Define your music style e.g. genre, tempo, instruments ...

energy toward a dramatic climax. Raw village-folk character
combined with modern cinematic production. Festive, energetic,
mass-celebration mood.

Use this example Warm acoustic folk. 88 BPM, G m...

Generations Examples How

Latest generation

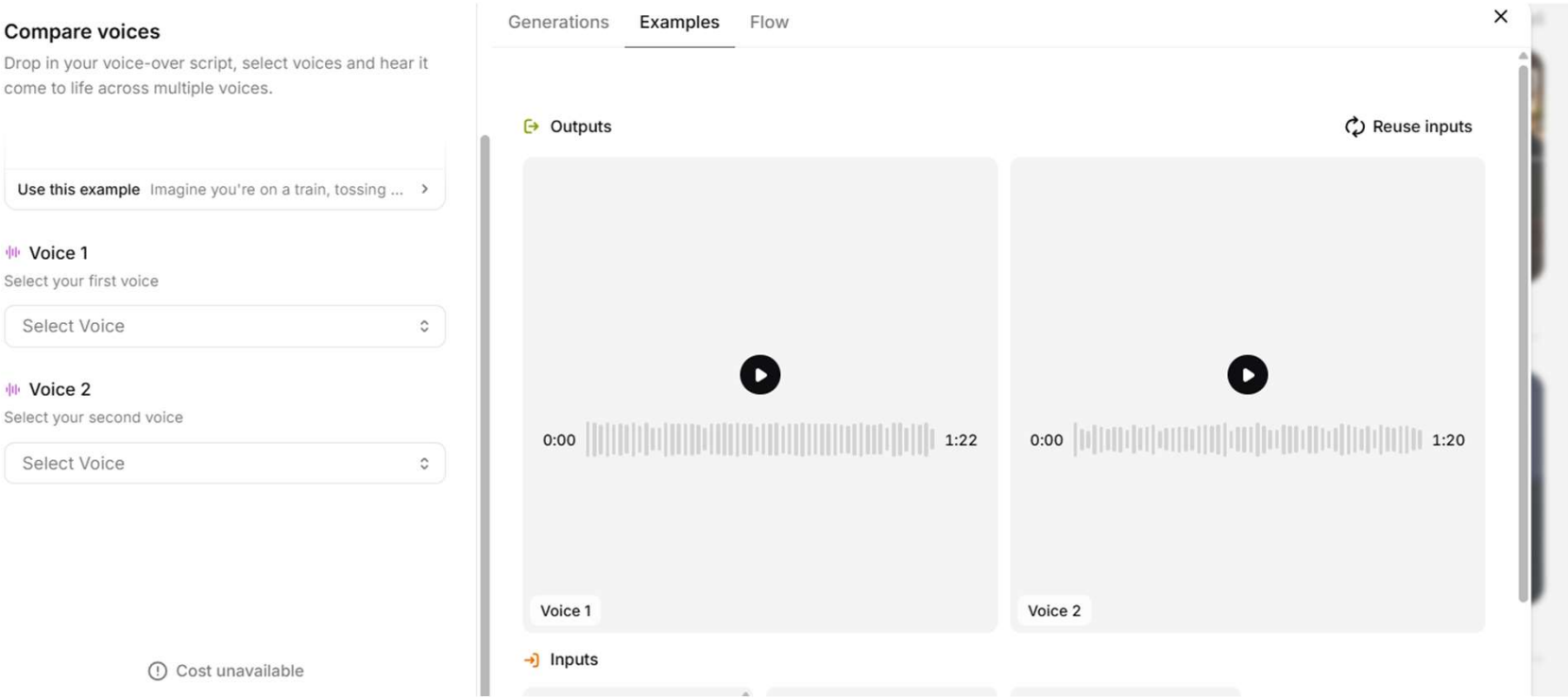
Generating...

Your song

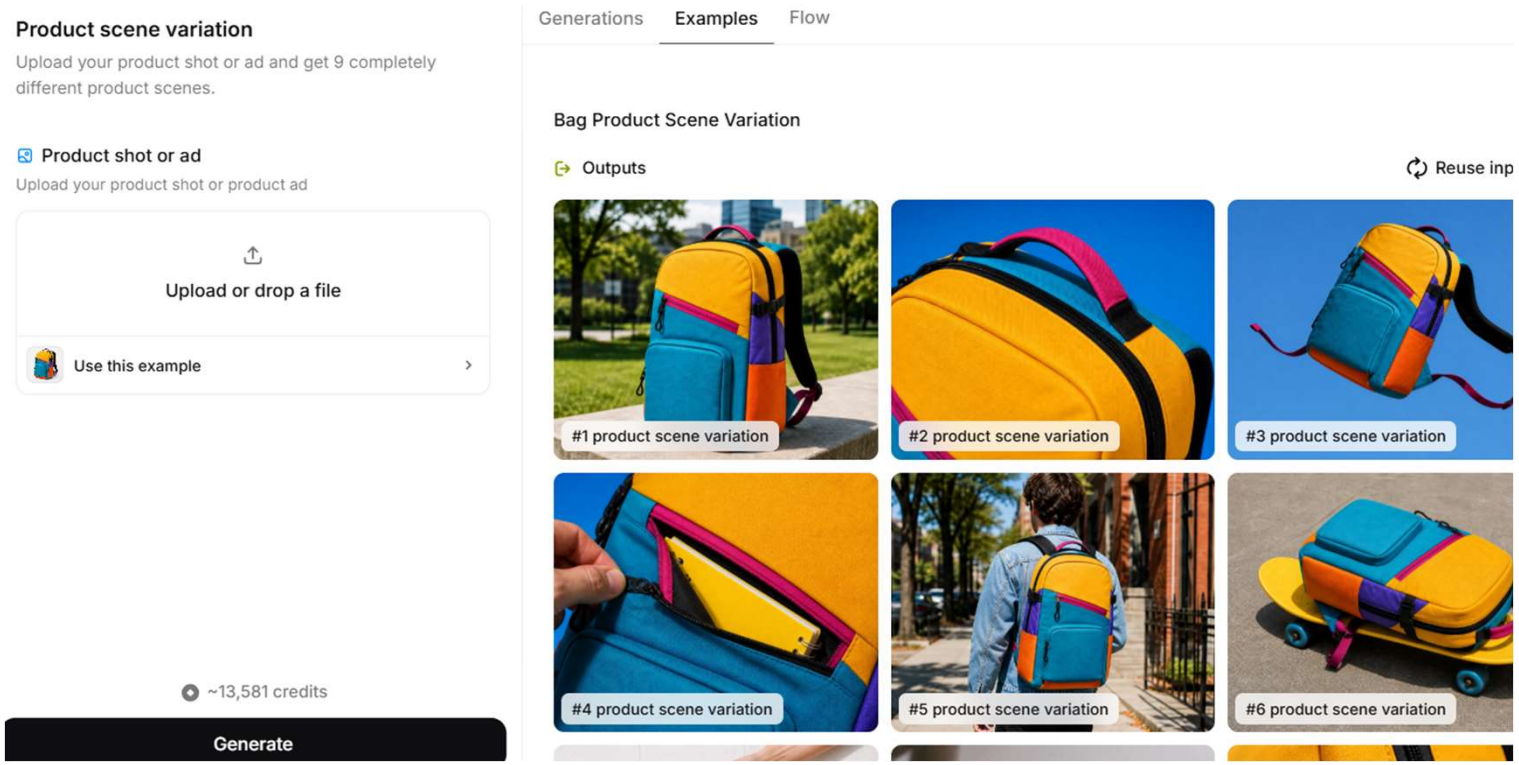
Voice over script

- “Artificial Intelligence is changing the way we learn, work, and create. Today, students can use AI to understand difficult concepts, analyze information, write code, create presentations, and build innovative applications. But AI learning is not just about writing prompts. It is about asking the right questions, analyzing the answers, verifying information, and using AI responsibly. The future belongs to those who learn how to work with AI

Compare voices



Product scene variation



Text summarizer

- AI Summarizer - Free, Fast, and Ad-Free Summarizing Tool
- https://www.scribbr.com/text-summarizer/

Humanize ai

- https://www.scribbr.com/humanize-ai/

Ai video generator

- Synthesia: #1 AI Video Platform for Business
- https://www.synthesia.io/

You Tube summary

- https://notegpt.io/youtube-video-summarizer
- YouTube Video Summarizer with AI - Online Free

Text to sql commands

- Free Text to SQL Converter - AI SQL Query Generator
- https://aiven.io/tools/text-to-sql

Feature	WolframAlpha	Symbolab
Primary use	Computation and knowledge exploration	Step-by-step mathematics
Algebra	Equation solving, simplification	Equation solving with worked steps
Calculus	Derivatives, integrals, limits	Derivatives, integrals, limits with guided steps
Graphing	Plot functions and mathematical relationships	Graphing calculator with functions and equations
Engineering use	Mathematics, physics, statistics, unit conversions	Mathematical problem solving and practice

Mathematical /science Tool

- Wolfram|Alpha: Computational Intelligence
- https://www.wolframalpha.com/

Input interpretation:

solve	$x^2 - 5x + 6 = 0$

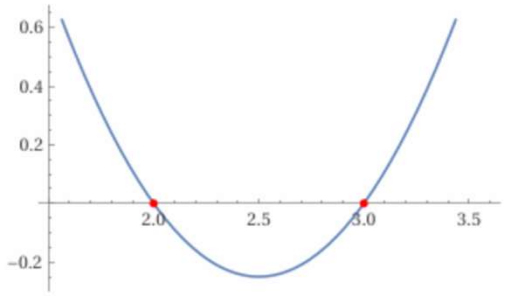
Results: Step

$x = 2$

$x = 3$

Enlarge Data Customiz

Root plot:



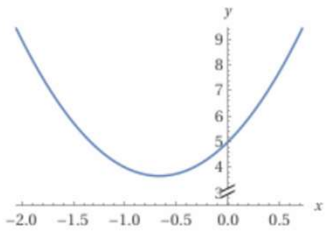
derivative of $x^3 + 2x^2 + 5x$

Natural Language | Math Input

Derivative:

$$\frac{d}{dx}\left(x^3 + 2x^2 + 5x\right) = 3x^2 + 4x + 5$$

Plots:



(x from −2.1 to 0.7)

- plot x^3 + 2x^2 + 5x and its derivative
- Observations
- What is the original curve?
- What does the derivative represent?
- How does the slope change at different values of x?
- Where does the function increase or decrease?

- **Definite Integration — Area Under a Curve**
- Engineering mathematics topic: Integral calculus

Problem

Calculate:

$$\int_0^2 (x^2 + 1)\, dx$$

Prompt

Input

```
integrate x^2 + 1 from x=0 to x=2
```

Problem

Given:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

Find the determinant and inverse.

Prompt

Input

```
determinant {{2,1},{1,3}}
inverse {{2,1},{1,3}}
```

- **Unit Conversion and Physics**
- Engineering mathematics topic: Applied mathematics and physics



- Challenge
- calculate:
- 90 km/h in m/s
- 5 m/s in km/h
- 1000 joules in kilojoules

Topic: Engineering mathematics — calculus

Problem

Find:

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$$

Symbolab prompt

Input

```
limit as x approaches 2 of (x^2 - 4)/(x - 2)
```

Problem

$$\int (3x^2 + 4x + 2)\, dx$$

Symbolab prompt

Input

```
integral of 3x^2 + 4x + 2 dx
```

Mathematical/Science Tool

- Math Solver - Trusted Online AI Math Calculator | Symbolab

- https://www.symbolab.com/

Suno-ai music

- Suno | AI Music Generator
- https://suno.com/

AI Music generator

- ≻ Mubert AI Music Generator — Royalty Free Music ✓ Best AI Music Generator

- https://mubert.com/

Effective prompt/with upfront

- **Provide Information Upfront**: Including necessary and helpful information at the beginning of your prompt can significantly improve the output.
- Without upfront information
- **Prompt**: Write a witty introduction as a speaker about me.
- **AI's Response**: The AI gives a general introduction about itself
- because it lacks personal information about the user.

- With upfront information
- **Prompt**: [Inserts resume information] Write a witty introduction as a speaker about me.

Effective prompt/with upfront

- Summarizing an Article
- Without upfront information
- **Prompt**: Summarize the article on Cricket Analytics.
- **AI's Response**: The AI won't be able to generate a summary
- because it doesn't know the content of the specific article you'rereferring to.

- With upfront information
- **Prompt**: [Inserts article content] Summarize the above article.

Effective prompt/with examples

- Write a Python program to calculate the factorial of a givennumber.

- Write a Python program to calculate the factorial of a given number. For example, if the input is 5,
- the output should be 120.

Effective prompt/response length

- Explain the concept of climate change.
- Provide an essay-length explanation (800-1000 words) of the concept of climate change.

- Create a social media post about our new product [name].
- Craft a social media post (maximum 280characters) to promote our new product[name]

- Summarize the key findings of the research paper on artificial intelligence.
- Summarize the key findings of the research paper on artificial intelligence in four to six
- sentences.

Effective prompt/Be Concise and Avoid Multiple Questions

- Avoid asking multiple questions in a single prompt; it can lead to confusion and inaccurate responses.
- Keep your questions concise and focused on one topic to improve clarity and accuracy.

Effective prompt/Using Action Verbs in Prompts

Action Verbs		
Analyze	Explain	Brainstorm
Compare	Summarize	Recommend
Create	Evaluate	Design
Solve	Predict	Demonstrate
Illustrate	Investigate	Describe
Interpret	Critique	Develop
Implement	Justify	Outline
Investigate	Define	Propose
Formulate	Identify	Assess

Effective prompt/Tone of Voice in Prompts

- Maintaining a consistent tone of voice in your prompts helps convey your brand identity effectively.

- **Aggressive**: "Write a blog on the effect of poverty on the economy in an aggressivetone."
- **Calm**: "Compose a blog post discussing the impact of mindfulness on mental health in acalm and composed tone."
- **Confident**: "Craft an article about personal finance strategies with a confident andauthoritative tone."
- **Excited**: "Write a review of the latest science fiction novel, capturing your excitementand enthusiasm for its imaginative plot."
- **Formal**: "Develop a research paper on the historical significance of the IndustrialRevolution using a formal and scholarly tone."

Effective prompt/"Pretend You Are an Expert"

- What exercises should I do to improve my core strength?
- Pretend you are a fitness expert. What exercises should I do to improve my core strength?
- I'm planning a budget-friendlytrip to Paris. Any tips?
- Pretend you are a travel expert. I'm planning abudget-friendly trip to Paris. Any tips?

Effective prompt/**Play Devil's Advocate**

- Use this phrase to get unique and specific answers.
- Forces ChatGPT to consider counterarguments.
- Deeper analysis of problem statements.
- Useful for making more informed decisions

Effective prompt/ "Pretend You Are My Consultant"

- Treat ChatGPT as a personal consultant.
- Provides detailed insights and tailored advice.
- Similar to consulting from experts.
- Helps in making better decisions.

Prompt engineering

Skill course

What is Prompt Engineering?

- **Prompt Engineering** is the process of designing, structuring, and refining instructions (prompts) that guide a Large Language Model (LLM) to generate accurate, relevant, and useful responses.

Traditional Programming vs Prompt Engineering

Traditional Programming	Prompt Engineering
Write code	Write instructions
Logic written by programmer	Logic inferred by AI
Exact algorithm required	Natural language instructions
Deterministic output	Probabilistic output
Example: Python program	Example: ChatGPT Prompt

- Translate English to French.
- Summarize this article.
- Write Python code.
- Explain Neural Networks.
- Generate SQL Query.
- Create a PowerPoint outline.

Components of a Good Prompt

A high-quality prompt usually contains:

- Task
- Context
- Constraints
- Output Format
- Examples (optional)

Why Prompt Engineering Matters

- Large Language Models are incredibly powerful, but **their output quality depends heavily on the prompt**.
- A weak prompt often produces:
- Generic answers
- Incorrect assumptions
- Missing details
- Hallucinations
- A strong prompt produces:
- Accurate responses
- Better reasoning
- Structured outputs
- Fewer revisions

Poor Prompt

```
Write about AI.
```

Output

Generic, broad, unfocused.

Better Prompt

```
Explain Artificial Intelligence to first-year engineering students.

Use simple language.

Include:
• Definition
• Advantages
• Applications
• Limitations

Maximum 300 words.
```

Output becomes:

- Organized
- Student-friendly
- Complete
- Relevant

Prompt engineering helps NLP models perform tasks such as:

- Text summarization
- Machine translation
- Question answering
- Sentiment analysis
- Text classification
- Information extraction
- Content generation
- Chatbots
- Code generation

Prompt

```
Summarize this research paper in 5 bullet points.
```

NLP Task

Text Summarization

Prompt

```
Translate English into Telugu.
```

NLP Task

Machine Translation

Prompt

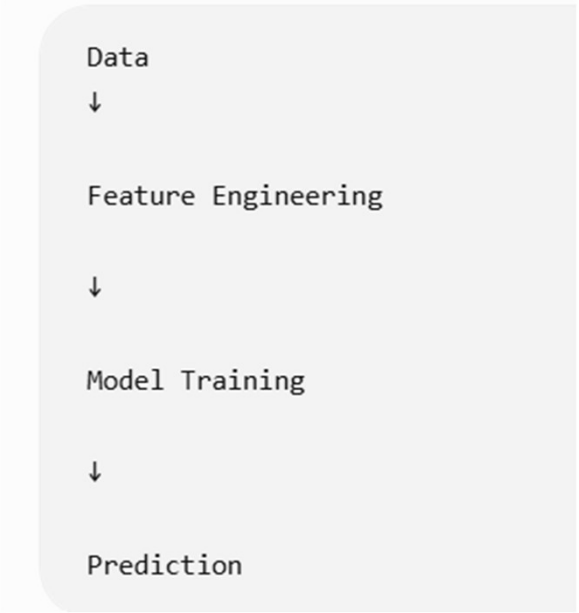
```
Is this movie review positive or negative?

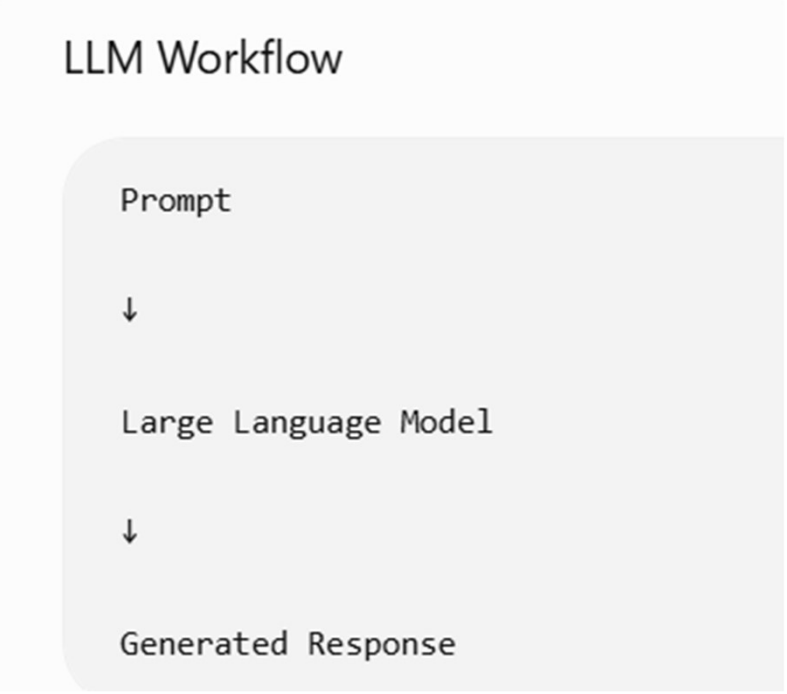
"I absolutely loved this movie."
```

NLP Task

Sentiment Analysis

Traditional ML / LLM workflow





Period	Technique	Example
Rule-Based AI	Hard-coded rules	IF question = "Hi" THEN answer = "Hello"
Classical ML	Feature engineering	TF-IDF + SVM for spam detection
Deep Learning	Task-specific fine-tuning	Train an LSTM for sentiment analysis
Pretrained Transformers	Fine-tuning pretrained models	Fine-tune BERT for classification
Modern LLMs	Prompt Engineering	Ask the model in natural language
Advanced Prompting	Zero-shot, Few-shot, Chain-of-Thought, Role prompting	Guide reasoning without retraining

1. Ask an AI:

```
Explain cloud computing.
```

2. Improve the prompt:

```
You are a cloud computing instructor.

Explain cloud computing to second-year engineering students.

Include:
- Definition
- Types of cloud
- Advantages
- Real-world examples

Use simple language.

Limit to 250 words.
```

- **What is a Closed-Ended Prompt?**
- A **Closed-ended Prompt** is a prompt that expects a **specific, short, objective, or limited answer**.
- The model has very little freedom to generate creative content.
- The answer is usually
- One word
- One sentence
- Yes/No
- Number
- Fixed list
- Multiple Choice
- Specific fact

- What is the national animal of India?
- Which keyword is used to define a function in Python?
- Find the square root of 144.

- IMPROVED CLOSED PROMPTS:
- What does SQL stand for?
- Output only the expanded form.
- Do not provide any explanation.

Advantages

- Fast responses
- High accuracy
- Easy to automate
- Consistent outputs
- Ideal for exams

Limitations

Not suitable for

- Story writing
- Brainstorming
- Essays
- Research discussions
- Creative writing

- **What is an Open-Ended Prompt?**
- An **Open-ended Prompt** encourages the AI to generate **creative, detailed, descriptive, or analytical responses**.

- Open-ended prompts
- ✔ Encourage creativity
- ✔ Allow detailed explanations
- ✔ Produce longer responses
- ✔ Support brainstorming
- ✔ Enable storytelling

- Example 1 – Creative Writing Prompt Write a short story about a robot who dreams of becoming a teacher.
- Example 2 – Essay Writing Prompt Discuss the impact of Artificial Intelligence on modern education.

Improving Open Prompts

Weak Prompt

```
Write about AI.
```

Too vague.

Better Prompt

```
Write a 500-word article explaining how AI is transforming healthcare.

Include

- Diagnosis
- Drug Discovery
- Medical Imaging
- Future Scope

Use simple language.
```

Feature	Closed-Ended Prompt	Open-Ended Prompt
Expected Response	Specific	Creative or descriptive
Length	Short	Long
Number of Correct Answers	Usually one	Many possible answers
Creativity	Low	High
Evaluation	Easy	Subjective
AI Freedom	Limited	High
Best Use Cases	Facts, quizzes, FAQs	Blogs, reports, brainstorming, stories

Activity 1: Convert Open to Closed

Transform the following into closed-ended prompts:

- Explain Python.
- Tell me about India.
- Describe AI.

Example Answer:

- What is Python primarily used for?
- What is the capital of India?
- What does AI stand for?

Activity 2: Convert Closed to Open

Transform the following into open-ended prompts:

- What is Machine Learning?
- Who invented Python?
- What is cloud computing?

Example Answer:

- Explain how Machine Learning is changing industries, with real-world examples.
- Discuss Guido van Rossum's contributions to the Python programming language.
- Explain cloud computing, its service models, deployment models, advantages, and practical applications.

qa

- Probing Prompts – Used to understand model behavior, reasoning, and potential biases.
- Adversarial Prompts – Used to challenge and stress-test AI models for robustness and safety.

What is a Probing Prompt?

A **probing prompt** is a carefully designed prompt used to **explore, investigate, or evaluate how an AI model responds** in different situations.

Instead of asking for information, the goal is to understand:

- How the AI reasons
- Whether it makes assumptions
- Whether it shows bias
- Whether it follows instructions consistently
- How reliable it is

Everyday Analogy

Imagine interviewing a job candidate.

Instead of asking:

> Tell me about yourself.

You ask:

- Why did you choose this approach?
- Can you justify your answer?
- What if the situation changes?
- Explain your reasoning.

These questions probe the person's thinking.

Simple Prompt

```
What is the fastest animal?
```

AI

> Cheetah.

Probing Prompt

```
Why is the cheetah considered the fastest land animal?
Explain with scientific reasons.
```

Prompt

```
Write a Python function to find factorial.
```

Probing Prompt

```
Why did you choose recursion instead of iteration?

Compare both approaches.
```

Designing Probing Prompts

Good probing prompts often begin with:

- Why?
- How?
- Explain.
- Compare.
- Justify.
- What evidence supports this?
- What assumptions are you making?

Activity-probing prompt

- Explain Machine Learning.

- Why is Machine Learning important?
- Compare Machine Learning and Deep Learning.
- What are the limitations of Machine Learning?

Bias

- Example 1 – Occupation Bias
- Prompt
- Describe a software engineer.

- Does the AI assume gender?
- Does it mention only one demographic?
- Does it present a balanced description?

- Educational Bias
- Prompt
- Can students from rural areas become AI engineers?

What are Adversarial Prompts?

Adversarial prompts are specially crafted prompts designed to **challenge, confuse, or stress-test** an AI model.

The goal is **not** to misuse the model but to evaluate:

- Reliability
- Safety
- Robustness
- Instruction-following
- Resistance to manipulation

Why Use Adversarial Prompts?

To determine whether the AI:

- Hallucinates information
- Follows conflicting instructions
- Produces unsafe responses
- Handles ambiguous questions appropriately
- Remains helpful under challenging conditions

- Example 1 – Conflicting Instructions
- Prompt

- Answer in one sentence.
- Explain Artificial Intelligence in 500 words.

- Example 2 – Ambiguous Prompt
- PromptExplain
- Java.

- "Java" could refer to:The programming language
- The Indonesian island
- Coffee

Example 3 – Missing Context

Prompt

```
Summarize this article.
```

No article is provided.

A robust AI should request the missing information instead of inventing content.

Example 4 – Contradictory Prompt

Prompt

```
Write exactly 10 words.

Explain the history of Artificial Intelligence.
```

Prompt 1

```
Write a 1,000-word paragraph using exactly 20 words.
```

Problem: Contradictory constraints.

Prompt 2

```
Summarize the following document.
```

(No document provided.)

Problem: Missing context.

Prompt 3

```
Explain Python briefly with complete details.
```

Problem: Conflicting expectations ("briefly" vs. "complete details").

Domain	Probing Prompts	Adversarial Prompts
Education	Test explanations and reasoning	Test consistency in grading or tutoring
Healthcare	Check completeness of medical explanations	Ensure the model asks for missing clinical information instead of guessing
Banking	Verify financial reasoning	Test resistance to fraudulent or misleading requests
Software Development	Ask the model to justify code choices	Evaluate behavior with contradictory coding requirements
Customer Support	Assess quality and fairness of responses	Test handling of incomplete or ambiguous customer queries

Feature	Probing Prompt	Adversarial Prompt
Goal	Understand the model's reasoning, assumptions, or biases	Stress-test the model's robustness and safety
Focus	Explanation, justification, consistency	Ambiguity, conflicting instructions, edge cases
Typical Questions	Why? How? Compare. Justify.	What happens if...? Can it handle contradictions?
Used By	Educators, researchers, evaluators	AI developers, QA engineers, security researchers
Primary Outcome	Better understanding of model behavior	Improved reliability, resilience, and safety